

Parul Sharma

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Objective

As a dedicated organic chemist specializing in synthetic organic and natural product chemistry, I have gained significant experience in natural product isolation during my master's and in organic synthesis during my PhD. Currently, I am in my fourth year, conducting research in Dr. Jimmie D. Weaver's Lab, working on my third project, which focuses on small molecule synthesis and developing new sustainable methods to synthesize useful bioactive compounds using visible light as a source of energy. My future goal is to work in an Industry where I can develop drug molecules for medical purposes.

Professional Preparation

PhD in Chemistry, Oklahoma State University (Advisor: Dr. Jimmie D. Weaver)	Expected May 2027
French Certificate, Punjab Agricultural University	2021
BS-MS Dual Degree (Chemistry), Punjab Agricultural University	2021

Appointments and Professional Experience

Graduate Research Assistant, Oklahoma State University	2023 – Present
Graduate Teaching Assistant, Oklahoma State University	2022 – 2023
Staff Manager, AS Components, Ludhiana, India	2021 – 2022
Chemistry Tutor (Self-employed)	2019 – 2022
Graduate Research Assistant, Punjab Agricultural University	2019 – 2021

Publications

- Hanumanthu, R., **Sharma, P.***, Ethridge, A., & Weaver, J. D. III.
Co-catalytic coupling of alkyl halides and alkenes: The curious role of lutidine.
J. Am. Chem. Soc., 2025, 147, 5238. [DOI]
- Kaur, A., Kaur, R., Bhardwaj, U., & **Sharma, P.**
Utilizing deep eutectic solvents for the green extraction of anticancer alkaloids from Catharanthus roseus.
Nat. Prod. Res., 2025, 1–9. [DOI]
- Kaur, R., Kaur, R., Singh, R., Bhardwaj, U., & **Sharma, P.**
Isolation and derivatization of bioactive compounds from Saussurea lappa roots with antifungal efficacy against plant pathogenic fungi.
J. Biol. Act. Prod. Nat., 2024, 14, 551. [DOI]
- Sharma, P.**, Kaur, R., Bhardwaj, U., & Kaur, J.
Chemical composition and antifungal potential of Vinca rosea leaf essential oil and extracts from Northern India.
Cogent Food Agric., 2024, 10, 2382317. [DOI]
- Sharma, P.** & Kaur, R.
A comprehensive review of Aloe vera: Composition, properties, processing, and applications.
Nat. Prod. J., 2025, 16. [DOI]
- Sharma, P.**, Kaur, R., & Kaur, A.
Antioxidant activity of deep eutectic solvent (DES) extracts of Catharanthus roseus.
Discov. Plants, 2025, 1, 242. [DOI]
- Sharma, P.**, Kaur, R., Kaur, J., & Bhardwaj, U.
Phytochemical profiling, antifungal potential of Vinca rosea extracts and isolated compound against wheat pathogens from Northern India.
Discov. Plants, 2024. [DOI]
- Sharma, P.** & Kaur, R.
Comprehensive analysis and therapeutic potential of ginger essential oil.
J. Pharmacogn. Phytochem., 2024, 13, 420. [DOI]
- Sharma, P.**, Singla, N., Kaur, R., & Bhardwaj, U.
A review on phytochemical constituents and pharmacological properties of Catharanthus roseus (L.) G. Don.
J. Med. Plants Stud., 2024, 12, 131. [DOI]

10. **Sharma, P.**
Pharmacological potentials and formulation strategies of Euphorbia hirta.
Eur. J. Med. Plants, 2024, 35, 63. [DOI]
11. **Sharma, P. & Kaur, R.**
Impact of chalcone on agriculture and its applications: An updated review.
Eur. J. Med. Plants, 2024, 35, 81. [DOI]
12. **Sharma, P.**
Exploring the bioactive constituents and medicinal properties of Hypericum androsaemum L.: A detailed review.
J. Med. Plants Stud., 2024, 12, 385. [DOI]
13. **Sharma, P. & Kaur, R.**
Unlocking the potential of chia seed: A comprehensive review of its potential applications.
Eur. J. Med. Plants, 2025, 36(1), 108–123. [DOI]
14. Kaur, R., **Sharma, P.**, Bhardwaj, U., & Kaur, R.
Dehydrocostus lactone: A comprehensive review on its isolation, chemical transformations, and pharmacological potential.
Discov. Chem., 2025, 1, 131. [DOI]
15. Kaur, N., Kaur, R., Bhardwaj, U., & **Sharma, P.**
Isolation and derivatization of bioactive compounds from Zingiber zerumbet essential oil with antifungal activity against rice pathogens.
Discover Plants. Preprint, 2025. [DOI]

Full list: Google Scholar

Academic and Professional Honors

Distinguished Graduate Fellowship in Chemistry, Oklahoma State University	2025
Merit Certificate for Highest CGPA in Chemistry, Punjab Agricultural University	2021
Poster Presentation Certificate of Participation	2019

Poster Presentations

GPCS Symposium, University of Oklahoma	Aug 2024
GPCS Symposium, Lawrence, Kansas	Sept 2023
ACS Southwest Regional Meeting, Waco, TX	Oct 2024
ACS Southwest Regional Meeting, Oklahoma City, OK	Nov 2023
Poster presented to Nobel Laureates, Punjab Agricultural University	Sept 2019
1st National Students' Conference on Spectroscopy (NCOS), GNDU Amritsar	Oct 2020

Professional Memberships

American Chemical Society	2023 – Present
Phi Lambda Upsilon Chemistry Honor Society	2023 – Present
Electrochemical Society (Social Media Officer)	2023 – Present

Courses Taught at OSU

CHEM 3013 — Survey of Organic Chemistry (TA), Assisted Dr Toby Nelson
CHEM 3012 — Organic Chemistry Laboratory

Licenses and Certifications

Programming for Everybody: Getting Started with Python (Coursera)	Nov 2020
Analytical Techniques in Molecules and Materials, SLIET	Jun 2020

Volunteered Services

Student Volunteer, National Service Scheme	2016 – 2019
Judge, Oklahoma State Science and Engineering Fair	Mar 2023
Volunteer, Grandparent University Event	Jun – Jul 2023
Member, Indian Student Association	Aug 2022 – Present
Contributor, Cancer Social Work	
Reviewer, Chemistry & Biodiversity Journal	2025 – Present